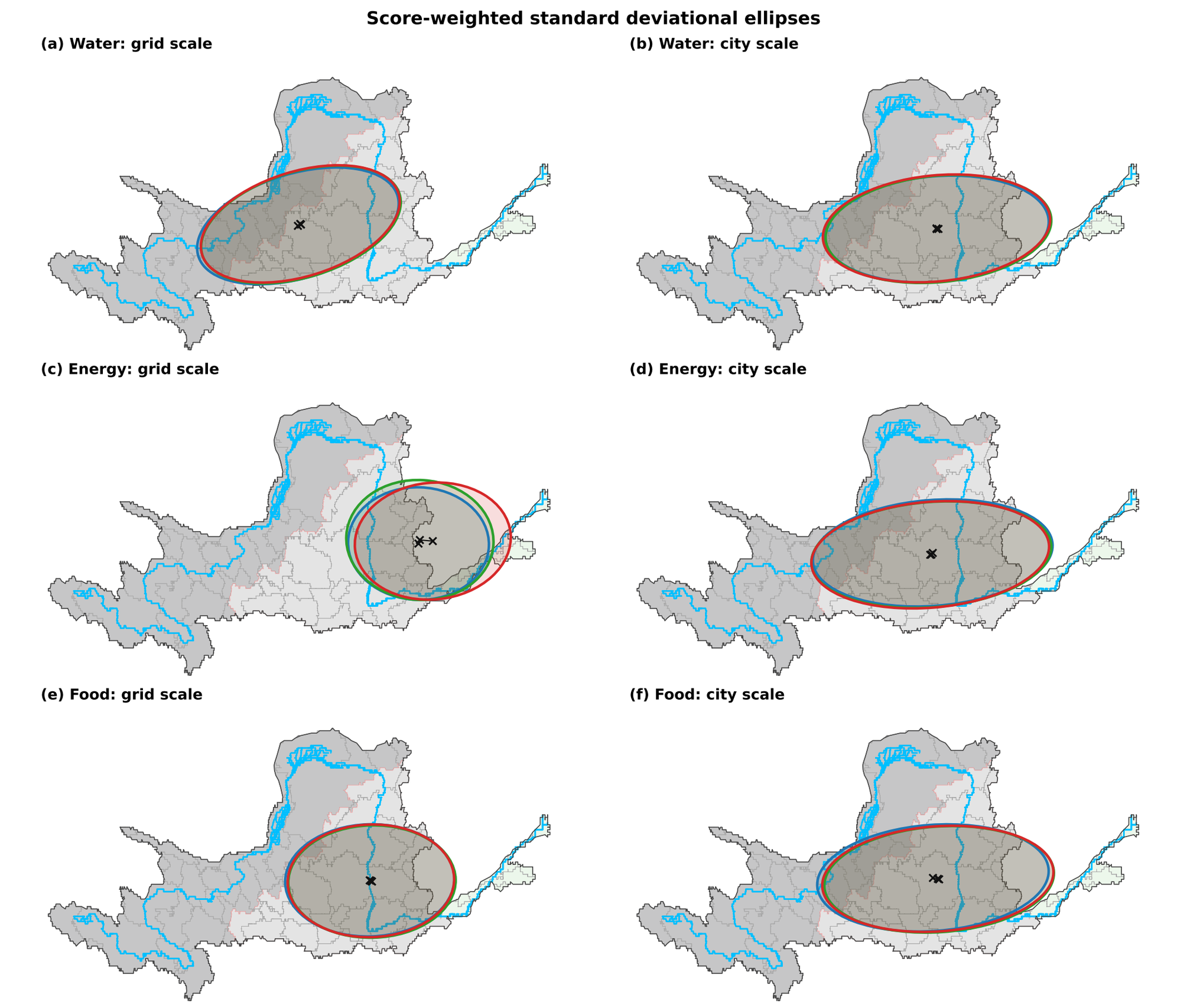

Score-weighted standard deviational ellipses
(a) Water: grid scale
(b) Water: city scale
(c) Energy: grid scale
(d) Energy: city scale
(e) Food: grid scale
(f) Food: city scale
(g) 2010–2020 centre displacement
Grid — Energy: 59.691 km E; Changzhi to Jinzhong | Food: 5.132 km SE; Yan'an to Linfen | Water: 12.470 km NE; Qingyang to Qingyang
City — Energy: 12.671 km SW; Yan'an to Yan'an | Food: 20.497 km E; Yan'an to Yan'an | Water: 27.60 km N; Yan'an to Yan'an
2010
2020
2015
Ellipse centre
Yellow River
City boundary
Upper reach
Middle reach
Lower reach